

Data Flow Diagram

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine (EALSE)
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through the Emotion-Aware Learning Support Engine. It represents the interaction between external entities, system processes, data stores, and the flow of information from a learner's text input to the generation of empathy-aware guidance. The diagram provides a high-level overview of how text is preprocessed, classified, scored, and finally presented back to the user.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded Shape	External Entity	A person, organisation, or external system that sends or receives information from the Emotion-Aware Learning Support Engine.
Rectangle with Number	Process	Represents a processing activity such as Text Preprocessing, Emotion Classification, or Guidance Generation.
Rectangle (Data Store)	Data Store	Stores information such as logged interactions and analytics data for future retrieval.
Labeled Arrow	Data Flow	Indicates the movement of information between external entities, processes, and data stores.

Data Flow Diagram Visualization

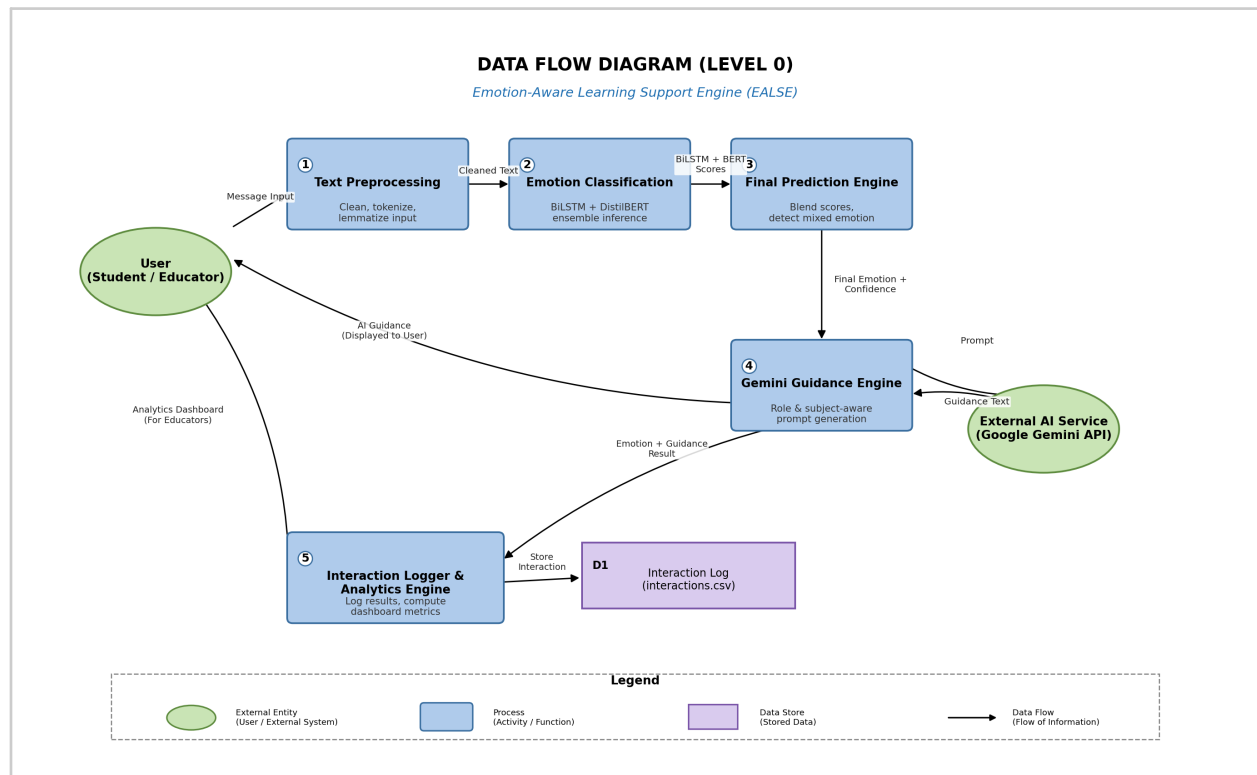


Figure 1: Data Flow Diagram of Emotion-Aware Learning Support Engine (EALSE)

Description

The workflow begins when the user (student or educator) enters a free-text message describing a learning struggle. The Text Preprocessing module cleans, tokenises, and lemmatises the input before forwarding it to the Emotion Classification Engine, which runs BiLSTM and DistilBERT inference in parallel. The Final Prediction Engine blends both models' scores, resolving ties and flagging mixed emotional states. This result is passed to the Gemini Guidance Engine, which builds a role- and subject-aware prompt and calls the external Google Gemini API to generate empathetic, personalised guidance. The generated guidance is displayed back to the user, while the Interaction Logger & Analytics Engine records the full result to the Interaction Log and powers the analytics dashboard used by educators to track emotional trends across sessions.